Exp-4: Create a visual representation in Figma highlighting the key elements of mobile network protocol stacks.

Aim:

Design a visual representation in Figma highlighting the key elements of mobile network protocol stacks.

Procedure:

1. open Figma

2. Create a new file

3. Select the Frames

4. Fill in the content that is required for presentation

5. Design Visual Elements

6. Make it Interactive

7. Add Annotations and Explanations

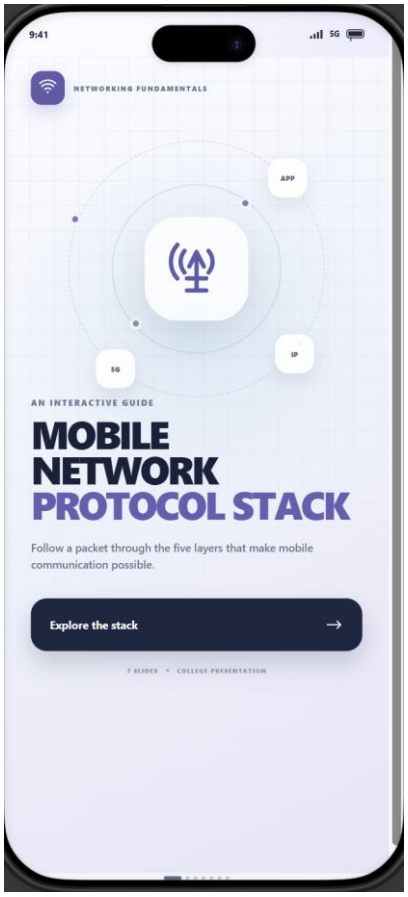
8. Incorporate Multimedia

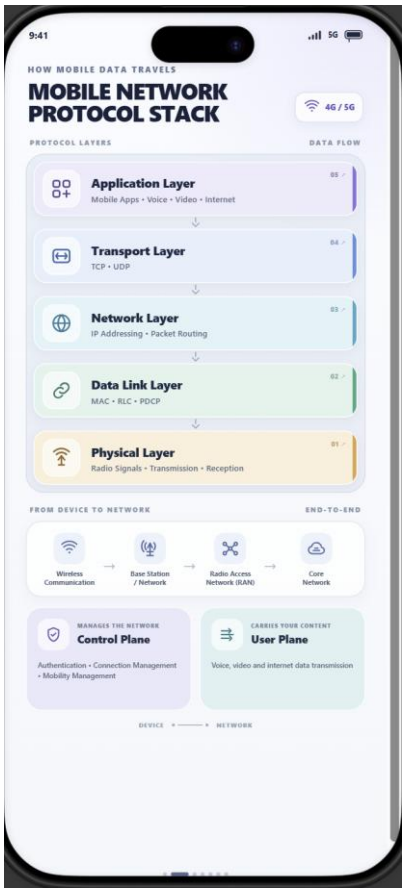
9. Storyboard Animation

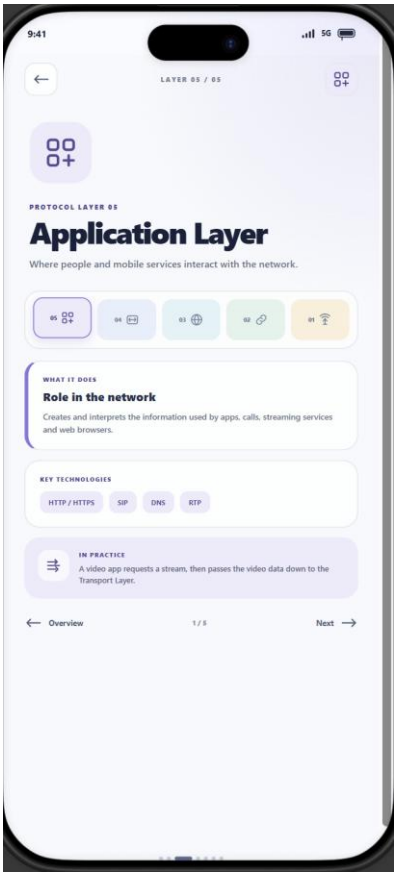
11. Review and edit the Prototype

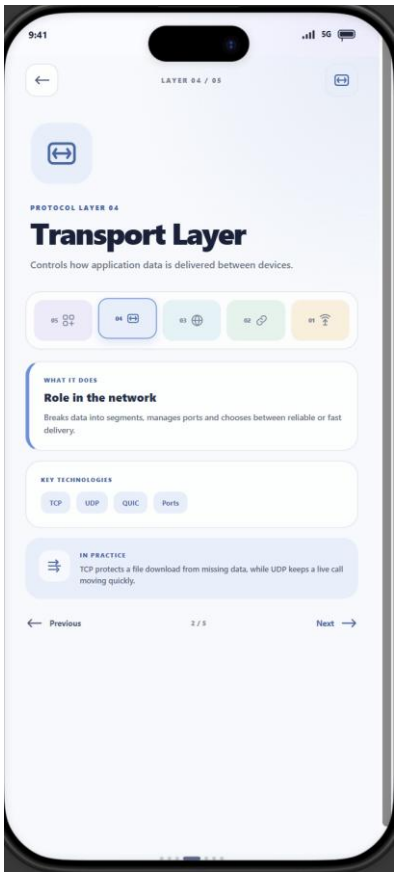
12. Save and Share

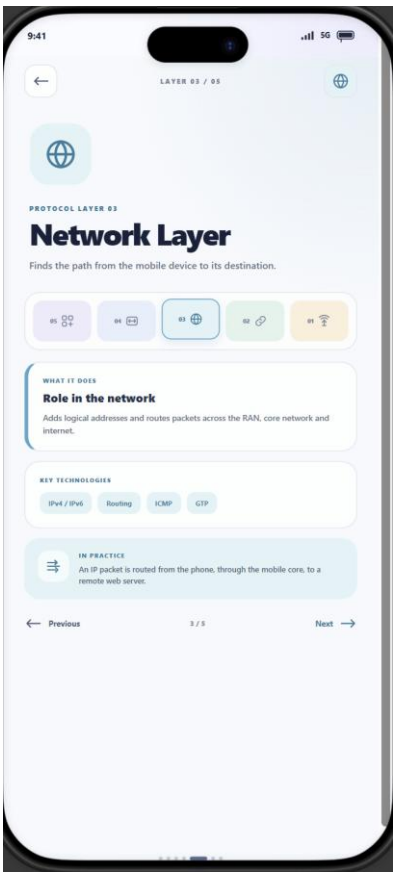
Design:

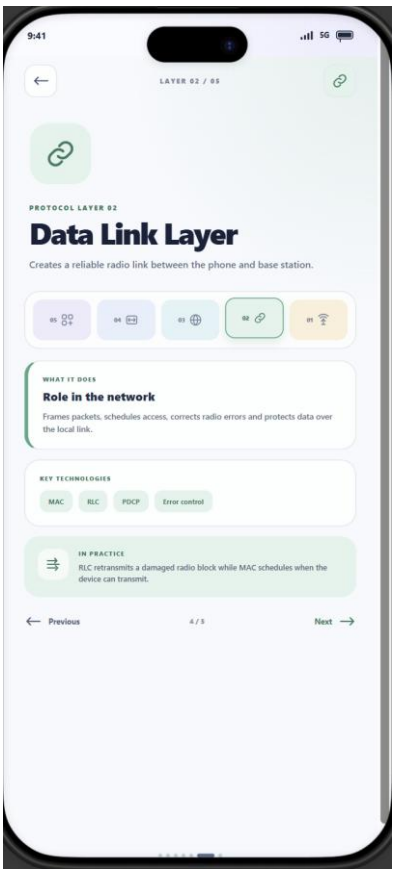

9:41
5G
NETWORKING FUNDAMENTALS
APP
IP
5G
AN INTERACTIVE GUIDE
MOBILE NETWORK PROTOCOL STACK
Follow a packet through the five layers that make mobile communication possible.
Explore the stack
7 SLIDES • COLLEGE PRESENTATION

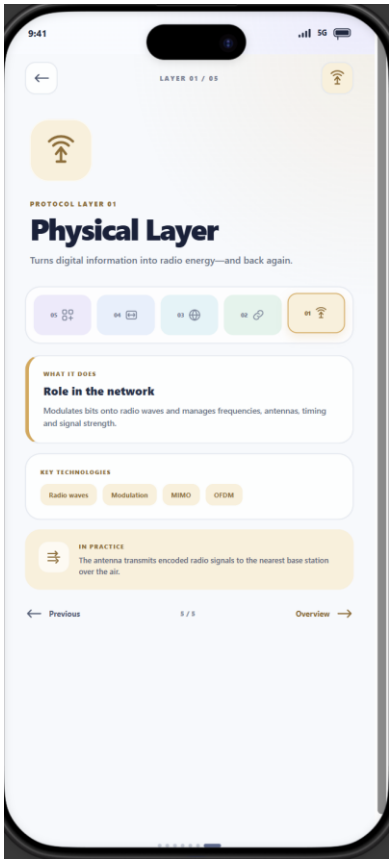

9:41
5G
HOW MOBILE DATA TRAVELS
MOBILE NETWORK PROTOCOL STACK
4G / 5G
PROTOCOL LAYERS
DATA FLOW
Application Layer
Mobile Apps • Voice • Video • Internet
05
Transport Layer
TCP • UDP
04
Network Layer
IP Addressing • Packet Routing
03
Data Link Layer
MAC • RLC • PDCP
02
Physical Layer
Radio Signals • Transmission • Reception
01
FROM DEVICE TO NETWORK
END-TO-END
Wireless Communication
Base Station / Network
Radio Access Network (RAN)
Core Network
MANAGES THE NETWORK
Control Plane
Authentication • Connection Management • Mobility Management
CARRIES YOUR CONTENT
User Plane
Voice, video and internet data transmission
DEVICE
NETWORK


9:41
5G
LAYER 05 / 05
PROTOCOL LAYER 05
Application Layer
Where people and mobile services interact with the network.
WHAT IT DOES
Role in the network
Creates and interprets the information used by apps, calls, streaming services and web browsers.
KEY TECHNOLOGIES
HTTP / HTTPS
SIP
DNS
RTP
IN PRACTICE
A video app requests a stream, then passes the video data down to the Transport Layer.
Overview
1 / 5
Next


9:41
5G
LAYER 04 / 05
PROTOCOL LAYER 04
Transport Layer
Controls how application data is delivered between devices.
WHAT IT DOES
Role in the network
Breaks data into segments, manages ports and chooses between reliable or fast delivery.
KEY TECHNOLOGIES
TCP
UDP
QUIC
Ports
IN PRACTICE
TCP protects a file download from missing data, while UDP keeps a live call moving quickly.
Previous
2 / 5
Next


9:41
5G
LAYER 03 / 05
PROTOCOL LAYER 03
Network Layer
Finds the path from the mobile device to its destination.
WHAT IT DOES
Role in the network
Adds logical addresses and routes packets across the RAN, core network and internet.
KEY TECHNOLOGIES
IPv4 / IPv6
Routing
ICMP
GTP
IN PRACTICE
An IP packet is routed from the phone, through the mobile core, to a remote web server.
Previous
3 / 5
Next


9:41
5G
LAYER 02 / 05
PROTOCOL LAYER 02
Data Link Layer
Creates a reliable radio link between the phone and base station.
WHAT IT DOES
Role in the network
Frames packets, schedules access, corrects radio errors and protects data over the local link.
KEY TECHNOLOGIES
MAC
RLC
PDCP
Error control
IN PRACTICE
RLC retransmits a damaged radio block while MAC schedules when the device can transmit.
Previous
4 / 5
Next


9:41
5G
LAYER 01 / 05
PROTOCOL LAYER 01
Physical Layer
Turns digital information into radio energy—and back again.
05
04
03
02
01
WHAT IT DOES
Role in the network
Modulates bits onto radio waves and manages frequencies, antennas, timing and signal strength.
KEY TECHNOLOGIES
Radio waves
Modulation
MIMO
OFDM
IN PRACTICE
The antenna transmits encoded radio signals to the nearest base station over the air.
Previous
5 / 5
Overview